17.7.4.9 Step 9: Add HTTPS with Let's Encrypt (Optional)

To secure your application with HTTPS, you can use **Certbot** to obtain a free SSL certificate from Let's Encrypt.

You can install Certbot and the Nginx plugin with the following commands:

```
sudo apt install certbot python3-certbot-nginx -y
sudo certbot --nginx -d yourdomain.com
```

Change `yourdomain.com` to your actual domain. Follow the prompts to set up SSL.

Certbot will automatically configure Nginx to use the SSL certificate.

17.7.5 Summary

In this lab, you've deployed a Laravel 12 app using **PHP 8.4** on an **Ubuntu Server VM** with Nginx and MySQL. This step-by-step process ensures a secure and production-ready Laravel deployment with the latest stable PHP features.

17.8 Exercise 51: Deploying Laravel 12 to Production in Docker on Ubuntu Server

17.8.1 Description

In this lab, you'll learn how to deploy a Laravel 12 application inside Docker containers on an Ubuntu Server. The application will run inside containers, but the MySQL database will remain on the host machine (outside of Docker). This hybrid setup is common in many production environments where databases are managed separately from app runtimes.

17.8.2 Objectives

By the end of this lab, you will be able to:

- Install and configure the latest Docker and Docker Compose on Ubuntu Server
- Create a containerized environment for Laravel 12 using PHP-FPM and Nginx